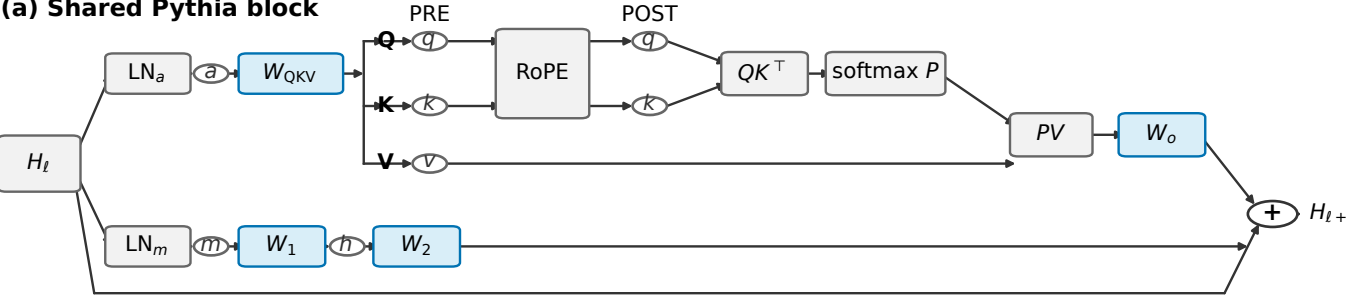


(a) Shared Pythia block



(b) Executed topologies; sites repeat in all six blocks

	branch			before RoPE		after RoPE		$V \rightarrow PV$	logical reach ceiling (%)
	a	m	h	Q	K	Q	K	V	$R_{\text{model}}^{\text{max}}$
A0	○	○	■	○	○	○	○	○	0
A1-H	○	○	●	○	○	○	○	○	4.2777
A3	●	●	●	○	○	○	○	○	11.7637
A4-Q	●	●	●	○	○	●	○	○	20.3233
A4-K	●	●	●	○	○	○	●	○	20.3233
A4-V	●	●	●	○	○	○	○	●	21.3928
A5-QK-PRE	●	●	●	●	●	○	○	○	20.3233
A5-QK-POST	●	●	●	○	○	●	●	○	20.3233
A6-PRE	●	●	●	●	●	○	○	●	29.9524
A6-POST	●	●	●	○	○	●	●	●	29.9524

● active gate ○ available port G = stock GELU at h operator, κ , optimizer, and pressure are separate; PRE/POST is Q/K only. Ceilings: all active outputs $\equiv 0$; logical product reach, not runtime. A6 closure: $V \equiv 0 \Rightarrow PV \equiv 0$; partial masks may not persist.